

JOLAPURAM

JAYAVARDAN

Embedded Software Engineer

Bluetooth Developer

CONTACT

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LinkedIn

GitHub

TECH SKILLS

- Languages
- Embedded C, Embedded C++,C++, Java
 - Python, MicroPython, Assembly
 - Arduino, Markdown
- Microcontrollers
- STM32F407VG, STM32F103C8
 - ARM7 LPC2129, PIC16F877A
 - ESP32, 8051, Raspberry Pi 4
 - TI LaunchPad F28p55xx
- Protocols
- SPI, I2C, UART, CAN
 - Bluetooth: HFP, A2DP, MAP, PBAP, OTA, PTS
 - HTTP, MQTT (IoT)
- Concepts
- Bare Metal, FreeRTOS,RTX RTOS
 - Linker Scripts, Startup Files
 - Timers, ADC, DAC, PWM
 - Interrupts, Custom Bootloader
 - ESP-IDF (CLI)

- Tools & Build
- GCC ARM, OpenOCD, GDB
 - Keil uVision, STM32CubeIDE
 - MPLAB X, ADB, SEGGER
 - Logic Analyser, Git/GitHub
 - Proteus, Cadence Virtuoso
 - Visual Studio, WinForms (GUI)
 - PlantUML (Diagrams)
- Platforms
- Linux/Unix System Prog.
 - Android (Bluetooth Stack)
 - ROS (Basics)
 - PCB Design, Soldering
 - LVGL (Embedded GUI)
 - Device Drivers (Linux/Embedded)
 - Linux Kernel Compilation & Customization

CERTIFICATIONS

- Embedded Systems — Eduskills & Microchip
- ARM Cortex, STM32 MCU1 & MCU2 — FastBit
- Cloud Computing — NPTEL (76%)
- IoT Intro — NPTEL (60%)
- Git & GitHub — GeeksforGeeks
- AI Model Dev — AIMERS Society

ACHIEVEMENTS

- 🏆 2nd Place — Circuit Debugging, Ripple 2K24
- 🏆 Appreciation — Chatbot Dev, AIMERS Society

PROFESSIONAL SUMMARY

ECE graduate with industry experience in Bluetooth protocol stack development (Android), embedded firmware, and automotive protocols. Currently at Harman International as a Bluetooth Developer. Proficient in bare-metal ARM Cortex-M4 programming, FreeRTOS, and communication protocols. Seeking an Embedded Software Engineer role.

EXPERIENCE

- Bluetooth Developer Intern · Harman International, Bengaluru Jan 2026 – Present
- Bluetooth stack development (Android) — hands-on HFP, A2DP, MAP, and PBAP profile implementation
 - OTA capture, Bluetooth PTS (Profile Tuning Suite) testing & certification workflows
 - Feature development and debugging in Java using ADB (Android Debug Bridge)
- Embedded Software Trainee · Vector India, Bengaluru Completed
- Interrupt-driven firmware and peripheral drivers for SPI, I2C, UART, CAN on ARM7 LPC2129
 - FreeRTOS task design, TCP/IP networking fundamentals, and Linux system programming
 - Strong grounding in memory management, interrupts, and peripheral interfacing
- ARM Cortex-M4 Bare Metal Programming · Argyan Tech Completed
- Cortex-M4 architecture deep-dive: registers, NVIC, exceptions, AAPCS, startup code, linker scripts
 - Built bare-metal task scheduler using SysTick & PendSV; debugged with GDB/OpenOCD
- PIC Microcontroller Course · Argyan Tech Completed
- Embedded C development on PIC16F877A — timers, ADC, UART, GPIO via register-level programming in MPLAB X (XC8)

TECHNICAL PROJECTS

- Custom Bootloader Development — STM32F407VG · Embedded C · UART · ARM Cortex-M4 2026
- UART-based custom bootloader — flash erase, memory write, CRC verification, and application jump functionality
 - VTOR relocation, linker scripts, startup code, and flash memory mapping for bootloader/app separation
 - Python host utility for firmware transfer and bootloader command handling over UART
- Bare-Metal Task Scheduler — STM32F407VG · GCC ARM CLI · OpenOCD · GDB 2026
- Built from scratch with NO HAL/IDE — custom linker scripts, startup files for precise memory layout
 - Cooperative/round-robin scheduling via SysTick & manual Cortex-M4 stack frame manipulation
 - Flashed and debugged via CLI; live register & stack frame inspection with GDB + OpenOCD
- Data Acquisition System — Vehicle Monitoring (RTOS) — ARM7 LPC2129 · FreeRTOS · SEGGER Sep–Oct 2025
- Real-time vehicle monitoring with FreeRTOS tasks for SPI, I2C, UART multi-sensor acquisition
 - SEGGER SystemView for real-time task profiling and RTOS trace visualization
- 32-bit Hybrid Ling & Ripple Carry Adder — Cadence Virtuoso Dec 2024–Mar 2025
- VLSI-designed 32-bit hybrid adder combining Ling and Ripple Carry logic for optimized speed and area
- Obstacle Detection & Line Following Robot — Arduino · Ultrasonic · IR · C++ — 2023
- Autonomous robot integrating ultrasonic and IR sensors for obstacle avoidance and line following
- AI Text and Speech Generation Model — Python · PyCharm — May–Jun 2024
- Python conversational AI model integrating text and speech I/O; recognized by AIMERS Society

EDUCATION

- B.Tech — Electronics & Communication Engineering
- Rajeev Gandhi Memorial College of Engg. & Tech., Nandyal | CGPA: 8.0/10 2022–2025
- Diploma — Electronics & Communication Engineering
- Sri Jyothi Polytechnic College, Vuyyuru | 89.2% 2019–2022
- Secondary Education (10th)
- ZPHS High School | GPA: 7.2/10 2019